

Bank Customer Churn Prediction and Retention Analytics Using Random Forest Machine Learning

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Abstract—Customer churn is one of the major challenges faced by banks and financial institutions. Losing existing customers affects revenue, customer lifetime value, and long-term business growth. This paper presents a machine learning based bank customer churn prediction system using the Random Forest Classifier. The dataset contains 10,000 customer records with demographic, financial, and behavioral attributes. Exploratory data analysis was performed to understand churn patterns across geography, gender, age group, product usage, and customer activity status. The proposed system uses data preprocessing, categorical encoding, model training, and evaluation to classify customers as churned or retained. The Random Forest Classifier achieved an accuracy of 84.2 percent, precision of 61.52 percent, recall of 59.71 percent, and F1-score of 60.6 percent. In addition, a Flask-based interactive dashboard and prediction system was developed and deployed on Render. The system helps identify high-risk customers and supports data-driven customer retention strategies.

Index Terms—Customer Churn, Machine Learning, Random Forest, Banking Analytics, Customer Retention, Flask, Business Intelligence

I. INTRODUCTION

Customer churn refers to the situation where existing customers stop using the services of an organization. In the banking sector, churn is a serious business issue because retaining existing customers is often more cost-effective than acquiring new customers. A bank may lose revenue, deposits, product usage, and future business opportunities when customers leave.

With the growth of digital banking, large amounts of customer data are generated through transactions, account activities, product usage, and customer profiles. Machine learning techniques can be used to analyze this data and identify hidden patterns related to customer behavior. By predicting churn in advance, banks can take proactive actions such as personalized offers, customer support improvement, loyalty programs, and targeted retention campaigns.

This paper focuses on developing a Bank Customer Churn Prediction System using the Random Forest Classifier. The system predicts whether a customer is likely to churn based on attributes such as credit score, age, geography, gender, tenure, account balance, number of products, credit card status, active membership status, and estimated salary.

II. PROBLEM STATEMENT

Banks often face difficulty in identifying customers who are likely to leave their services. Traditional reporting methods may show historical churn trends, but they may not effectively

predict future churn risk. The objective of this project is to build a machine learning system that can predict customer churn and provide business insights for customer retention.

The major goals are:

- To identify customers likely to churn.
- To analyze key churn drivers.
- To evaluate model performance using classification metrics.
- To deploy the model as a web-based prediction system.
- To provide dashboard-based business insights.

III. DATASET DESCRIPTION

The dataset used in this project contains 10,000 customer records from the banking domain. The target variable is *Exited*, where 1 represents a churned customer and 0 represents a retained customer.

TABLE I
DATASET SUMMARY

Parameter	Value
Total Customers	10,000
Churned Customers	2,037
Retained Customers	7,963
Churn Rate	20.37%
Retention Rate	79.63%
Average Credit Score	650.53
Average Balance	76,485.89

TABLE II
IMPORTANT DATASET FEATURES

Feature	Description
CreditScore	Customer credit score
Geography	Customer country or region
Gender	Customer gender
Age	Age of the customer
Tenure	Years associated with bank
Balance	Customer account balance
NumOfProducts	Number of bank products used
HasCrCard	Credit card ownership status
IsActiveMember	Customer activity status
EstimatedSalary	Estimated salary of customer
Exited	Churn status

IV. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to understand churn distribution and customer behavior patterns. The overall

churn rate was found to be 20.37 percent, which means approximately one out of five customers left the bank.

Country-wise analysis showed that customers from Germany had a higher churn tendency compared to customers from France and Spain. Gender-wise analysis indicated that female customers had slightly higher churn tendency than male customers. Age group analysis showed that middle-aged and older customers were more likely to churn compared to younger customers.

Product usage analysis showed that customers using fewer banking products were more likely to leave. Customers who were inactive members also showed higher churn risk. These findings indicate that customer engagement and product usage play an important role in retention.

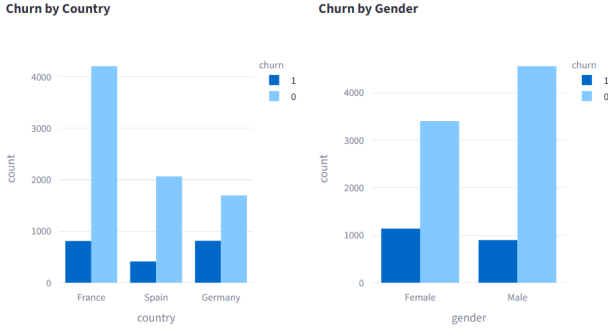


Fig. 1. Country-wise Customer Churn Analysis

V. METHODOLOGY

The proposed system follows a complete machine learning pipeline consisting of data preprocessing, feature selection, model training, evaluation, and deployment.

A. Data Preprocessing

Unnecessary columns such as customer identifier and surname were removed because they do not contribute to prediction. Categorical variables such as Geography and Gender were encoded into numerical format. The dataset was then prepared for training and testing.

B. Feature Selection

The selected input features include CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. The output variable is Exited.

C. Model Training

The Random Forest Classifier was used as the final prediction model. Random Forest is an ensemble learning algorithm that combines multiple decision trees and produces a final prediction using majority voting. It is effective for structured tabular datasets and helps reduce overfitting.

D. Model Evaluation

The trained model was evaluated using accuracy, precision, recall, F1-score, and confusion matrix. These metrics help measure the model's ability to correctly classify churned and retained customers.

VI. EVALUATION METRICS

The performance of the model was measured using the following formulas:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 \text{ Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Here, TP represents True Positive, TN represents True Negative, FP represents False Positive, and FN represents False Negative.

VII. RESULTS AND DISCUSSION

The Random Forest Classifier achieved good performance on the customer churn dataset. The overall accuracy was 84.2 percent. The precision was 61.52 percent, recall was 59.71 percent, and F1-score was 60.6 percent.

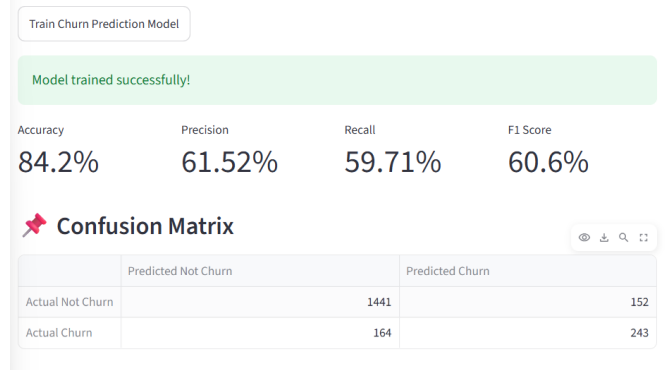


Fig. 2. Confusion Matrix of the Random Forest Classifier showing the classification performance on churn prediction.

TABLE III
MODEL PERFORMANCE

Metric	Value
Accuracy	84.2%
Precision	61.52%
Recall	59.71%
F1 Score	60.6%

The results show that the model can effectively classify customers into churned and retained categories. The confusion matrix helped in understanding correct and incorrect predictions. Since churn prediction is a business-sensitive problem, recall is important because banks need to identify as many churn-risk customers as possible.

VIII. DASHBOARD AND DEPLOYMENT

A Flask-based web application was developed to make the system practical and accessible. The application allows users to enter customer details and receive churn prediction results. The dashboard also displays key performance indicators and visual analytics.

The dashboard includes:

- Total customers, churned customers, and retained customers.
- Churn rate and retention rate.
- Country-wise churn analysis.
- Gender-wise churn analysis.
- Age group churn analysis.
- Product usage analysis.
- Confusion matrix visualization.
- Business insights.

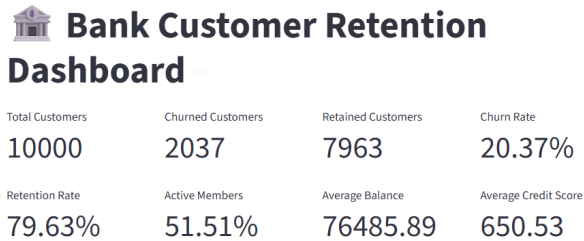


Fig. 3. Interactive Bank Customer Retention Dashboard

Low Churn Risk: 13.5%

Fig. 4. Customer churn risk prediction interface showing the predicted churn probability generated by the deployed Random Forest model.

The deployed web application provides real-time churn prediction capabilities for banking customers. Based on customer attributes such as credit score, age, balance, tenure, and activity status, the system estimates the probability of customer churn. The example shown in Fig. 4 predicts a churn risk of 13.5%, indicating a low likelihood of customer attrition. Such predictions can assist banks in identifying at-risk customers and implementing targeted retention strategies.

The application was deployed on Render, which makes the model accessible through a live web interface.

IX. BUSINESS IMPACT

The proposed system can help banks identify high-risk customers before they leave. It supports customer retention teams in designing targeted strategies. For example, inactive customers, customers with fewer products, and customers from high-churn regions can be prioritized for personalized offers or service improvements.

This type of predictive analytics system can improve customer satisfaction, reduce churn rate, and support better business decision-making.

X. CONCLUSION

This paper presented a bank customer churn prediction system using the Random Forest Classifier. The system analyzed 10,000 customer records and achieved an accuracy of 84.2 percent. Exploratory analysis showed that geography, age, product usage, and activity status influence churn behavior.

The integration of a Flask-based dashboard and Render deployment makes the project an end-to-end machine learning solution. The system can help banks detect churn-risk customers and take proactive retention actions.

XI. FUTURE SCOPE

Future improvements may include:

- Using advanced models such as XGBoost and LightGBM.
- Applying hyperparameter tuning for better performance.
- Adding explainable AI techniques such as SHAP or LIME.
- Integrating real-time customer data.
- Building automated retention recommendation systems.

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